



Introduction to Bioengineering Theory

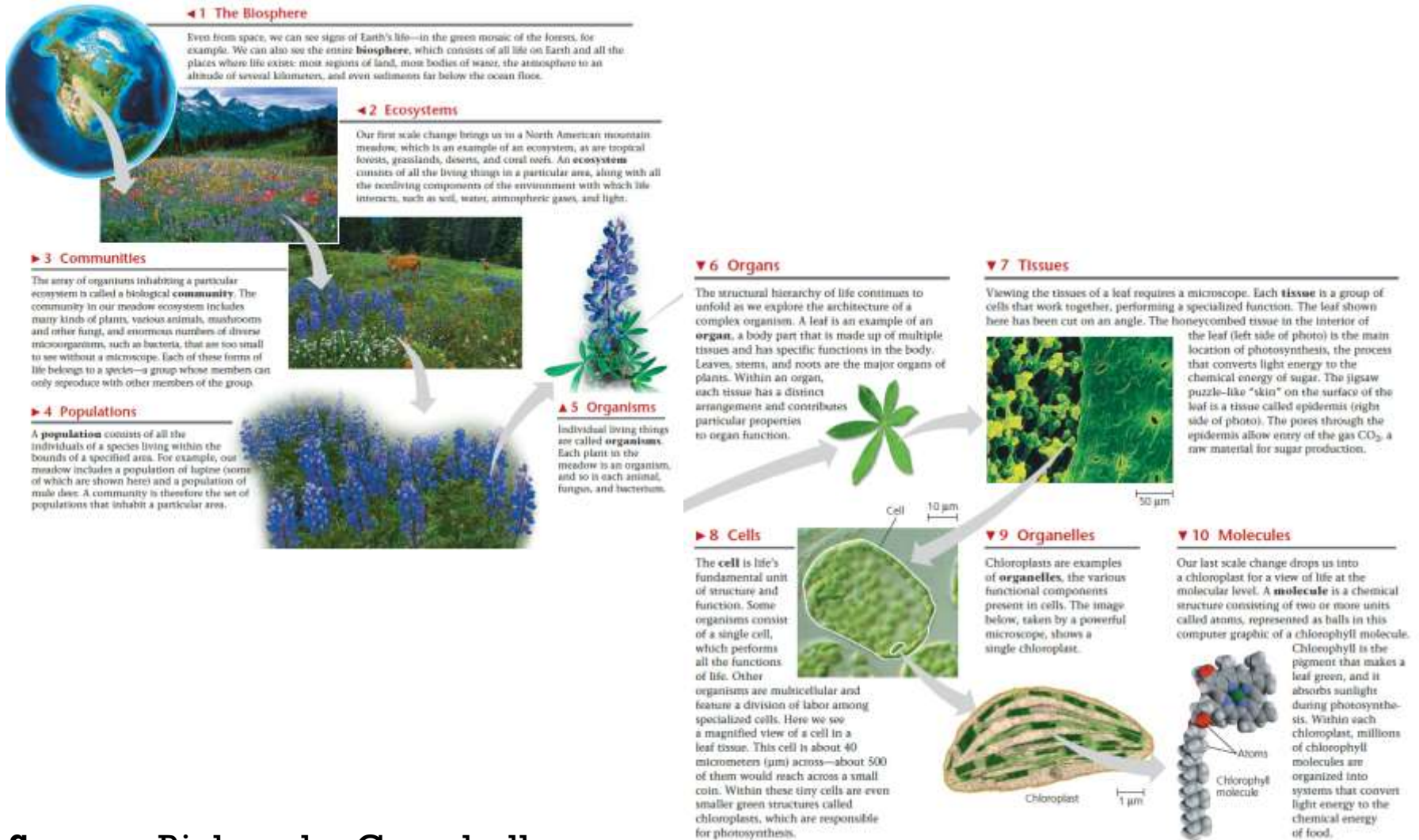
(BBL1020; 3-0-0)

Lecture 6

Today's topic

Cell Theory

Exploring levels of Biological Organization

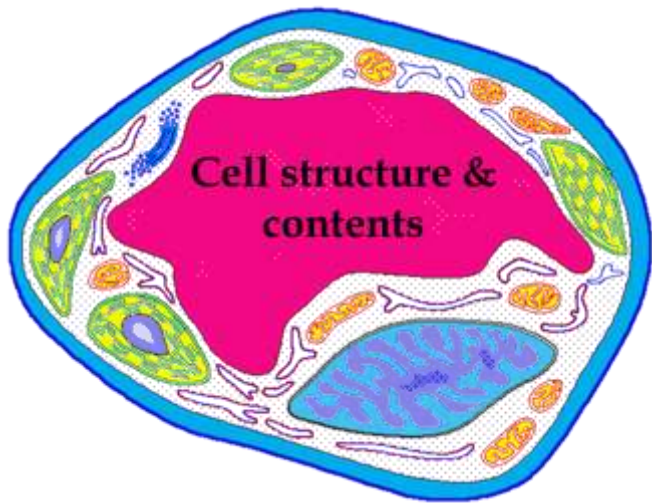


Source: Biology by Campbell;

Chapter-1: Evolution, the Themes of Biology, and Scientific Inquiry

Cell Theory

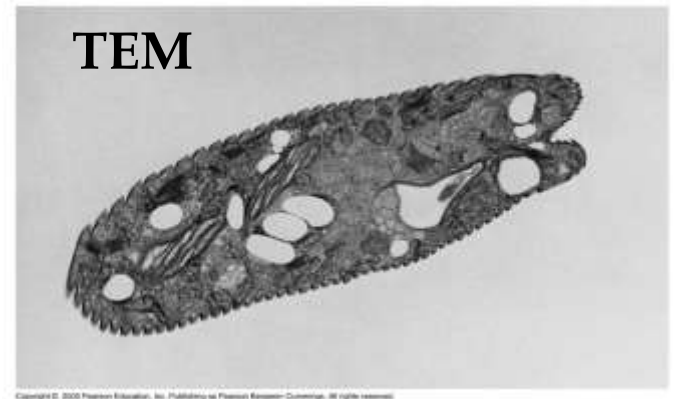
- Cells are the smallest living things. All organisms are composed of cells. They arise only from pre-existing cells.
- Cell size is limited – as the cell size increases, it takes longer for a material to diffuse from the cell membrane to the interior of the cell. As the cell size is increased, the volume increases 10 times faster than the surface area. Light and electron microscopes can aid visualization of cells.



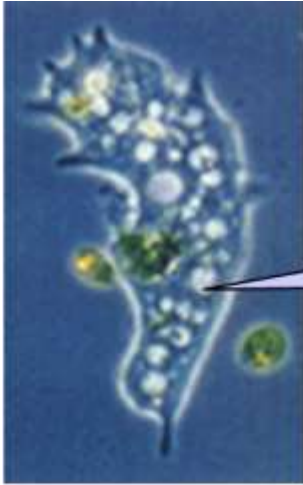
Cell radius (r)	1 unit	10 unit
Surface area ($4\pi r^2$)	12.57 unit ²	1257 unit ²
Volume ($\frac{4}{3}\pi r^3$)	4.189 unit ³	4189 unit ³
Surface Area / Volume	3	0.3

Cell Theory

- All cells have genetic material – in a nucleoid or nucleus, cytoplasm – a semifluid matrix, and plasma membrane - phospholipid bilayer.
- A cell is the smallest unit that can perform life functions.
- Light microscope - Can observe living cells in true color; Magnification of up to $\sim 1000\times$; Resolution ~ 0.2 microns – 0.5 microns.
- Electron microscopes - Magnification up to $\sim 100,000\times$; Transmission electron microscope (TEM) - 2-D image; Scanning electron microscope (SEM) - 3-D image.
Resolution ~ 0.1 nm – 0.5 nm.



Examples of Cells



Amoeba Proteus



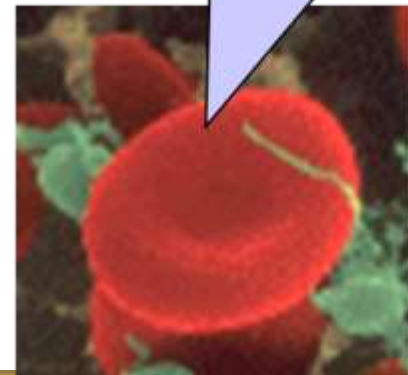
Plant Stem

Bacteria



Nerve Cell

Red Blood Cell

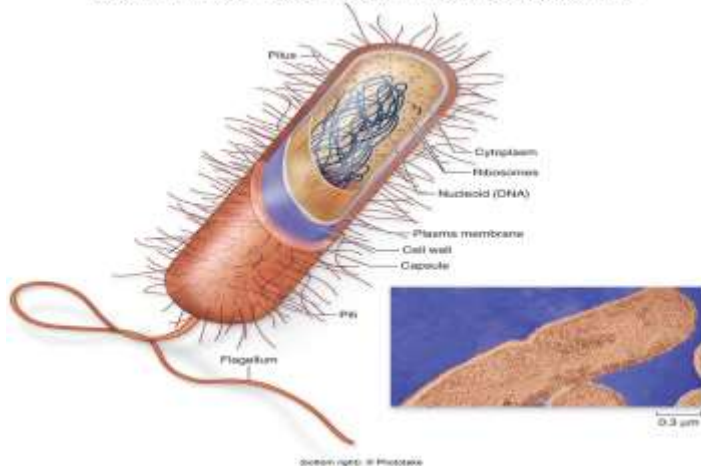


Cell Types

Prokaryotic

- Do not have structures surrounded by membranes
- Few internal structures
- Single-cell organisms, Bacteria
- Prokaryotic cells lack a membrane-bound nucleus - genetic material is present in the nucleoid
- Archaea and bacteria (types of prokaryotes).

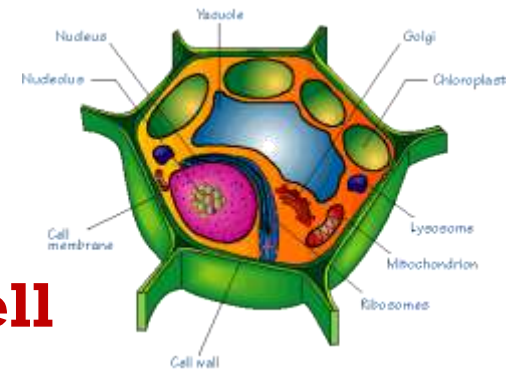
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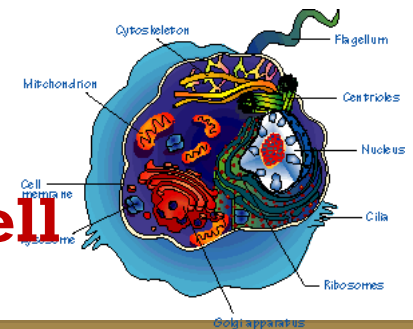
Eukaryotic

- Contain organelles
- Contain membrane bound nucleus, cytoskeleton for support and to maintain cellular structure.

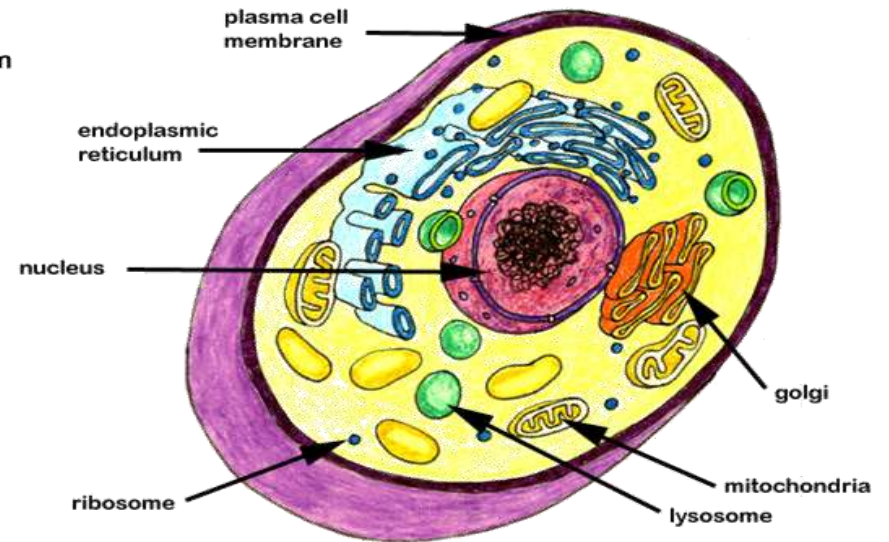
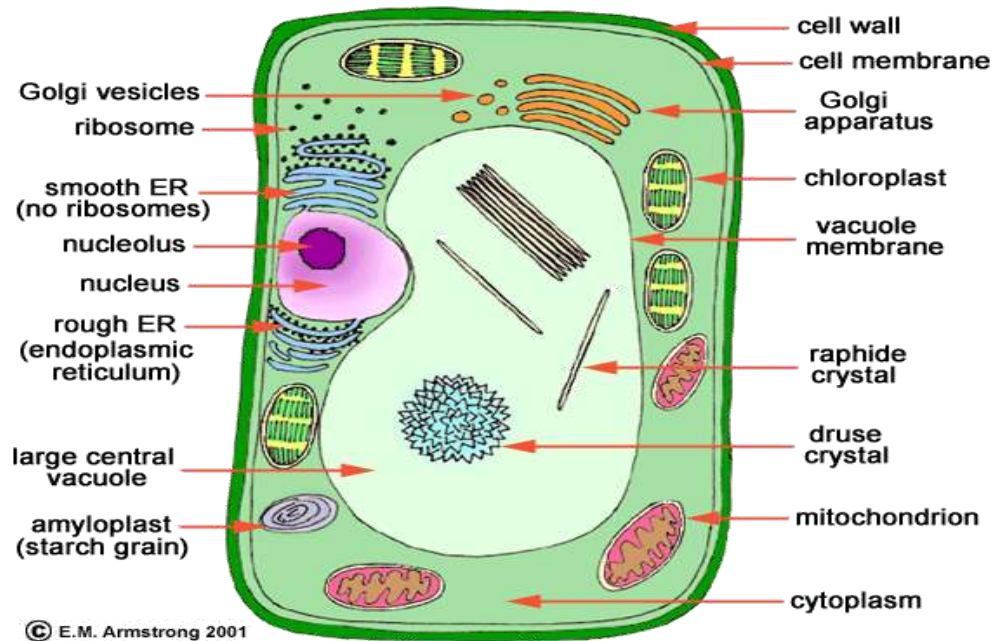
Plant cell



Animal cell

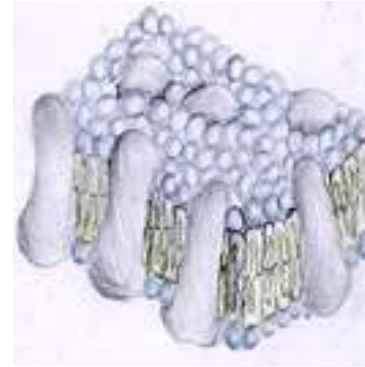


Plant and Animal Cell

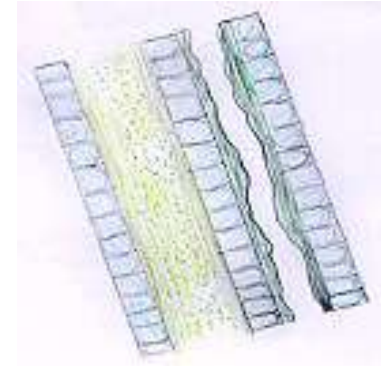


Functions of cell

- **Cell membrane** – Outer membrane of cell that controls movement in- and out-of-the cell. It is made up of a phospholipid bilayer. It contains many types of membrane proteins.
- **Cell wall** – Mostly found in plant cells and bacteria. It supports and protects cells.
- **Endoplasmic reticulum** - Moves materials around in cell. Smooth type: lacks ribosomes; Rough type (pictured): ribosomes embedded in surface.



Cell membrane



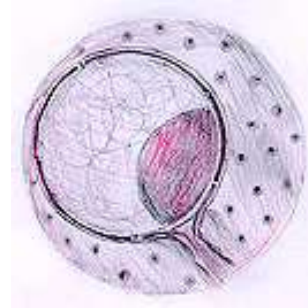
Cell wall



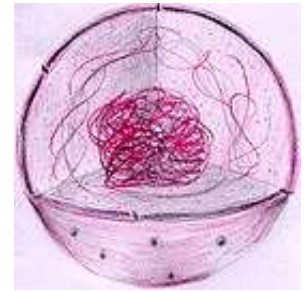
**Endoplasmic
reticulum**

Functions of cell

- **Nucleus** - stores the genetic material of the cell in the form of multiple, linear chromosomes. It is surrounded by a nuclear envelope composed of 2 phospholipid bilayers. In chromosomes, DNA is organized with proteins to form chromatin.
- **Nuclear membrane** – Surrounds nucleus; made of two layers; openings allow material to enter and leave nucleus.
- **Nucleolus** – Inside nucleus; contains RNA to build proteins.
- **Ribosomes** - the site of protein synthesis in the cell. It is composed of ribosomal RNA and proteins. It is found within the cytosol of the cytoplasm and attached to internal membranes.



**Nuclear
membrane**



Nucleolus



Ribosomes

Functions of cell

- **Mitochondria** – Powerhouse of the cell; produces energy through chemical reactions – breaking down fats & carbohydrates; Controls level of water and other materials in cell; Recycles and decomposes proteins, fats, and carbohydrates. It is filamentous or globular in shape.
- **Golgi bodies** – Protein 'packaging plant'; Move materials within the cell; Move materials out of the cell.
- **Lysosome** – Digestive 'plant' for proteins, fats, and carbohydrates; Transports undigested material to cell membrane for removal; Cell breaks down if lysosome structure is disrupted.
- **Vacuoles** - Membrane-bound sacs for storage, digestion, and waste removal; Contains water solution; Help plants maintain shape.



Mitochondria



Golgi bodies



Lysosome



Vacuoles

Summary

Components

- **Plasma membrane**
- **Cytosol**
- **Mitochondria**
- **Endoplasmic reticulum**
- **Golgi apparatus**
- **Lysosomes**
- **Cytoskeleton**
- **Nucleus**

Major functions

- **Transport of ions and molecules.**
- **Metabolism of carbohydrates, lipids and amino acids**
- **Energy production**
- **Synthesis of proteins and lipids**
- **Modification and sorting of proteins**
- **Cellular digestion**
- **Cell morphology and cell motility**
- **DNA synthesis and repair**

Summary

Overview of Cell Structure



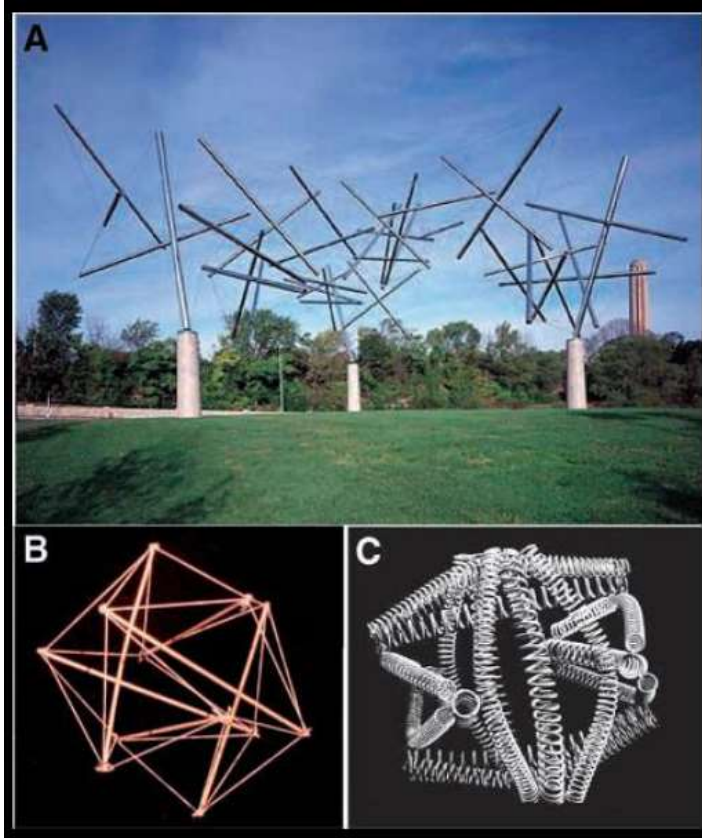
<https://youtu.be/0xe1s65IH0w?si=hLtQZMNHWMwqzOLd>

Endosymbiotic theory

https://www.youtube.com/watch?v=IhF5G2k45vY&ab_channel=PBSEons

Tensegrity

https://www.youtube.com/watch?v=0onncd0_0-o&ab_channel=SteveMould



Tensegrity: building principle, first described by R. Buckminster Fuller (architect) structures that stabilize their shape by continuous tension or 'tensional integrity' rather than by continuous compression.

Tensegrity structures:

(A) Triple crown, a tensegrity sculpture, that is composed of stainless-steel bars and tension cables.

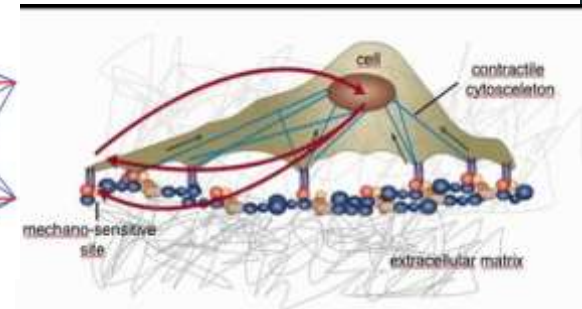
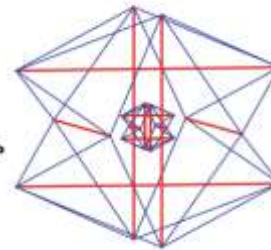
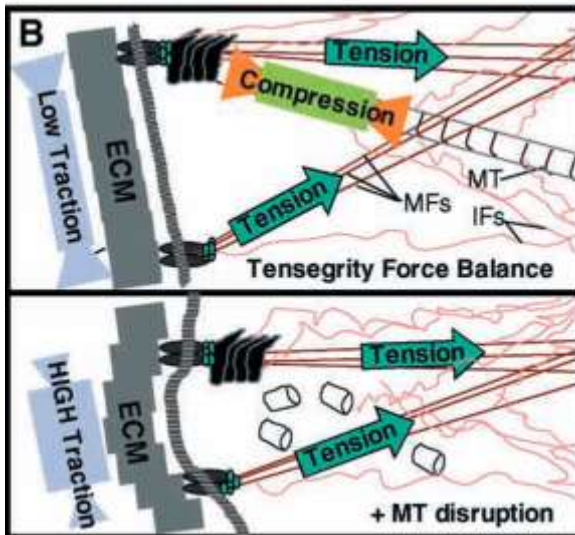
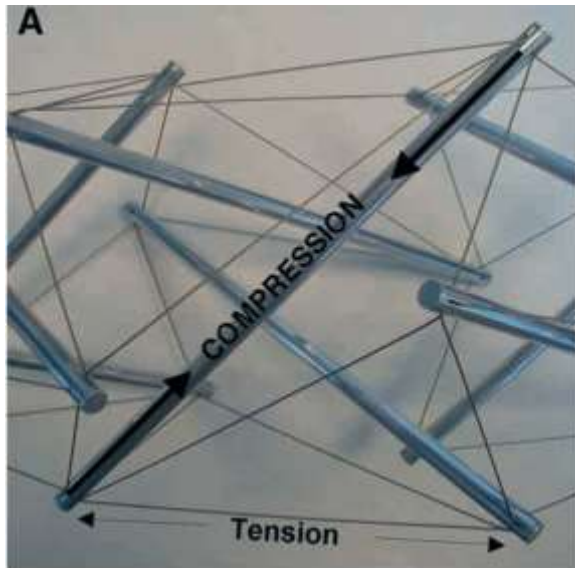
(B) A tensegrity sphere composed of six wood struts and 24 white elastic strings.

(C) The same tensegrity configuration as in B constructed entirely from springs with different elasticity.

[doi:10.1242/jcs.00359](https://doi.org/10.1242/jcs.00359)

Tensegrity is used at multiple size scales in the hierarchy of life — from individual molecules to whole living organisms — to both stabilize three-dimensional form and to channel forces from the macroscale to the nanoscale, thereby facilitating mechanochemical conversion at the molecular level.

Cellular Tensegrity Model

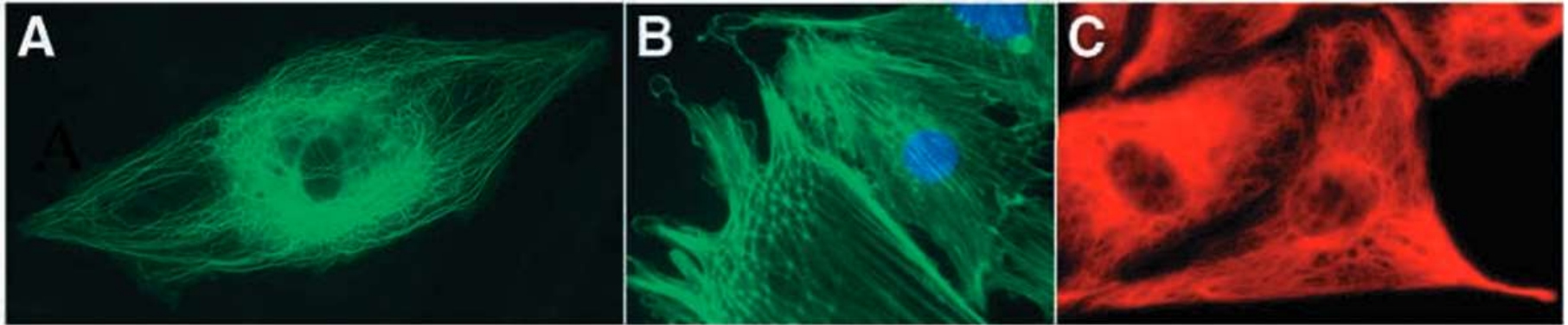


- Tensional forces are borne by cytoskeletal microfilaments and intermediate filaments.
- Balanced by interconnected structural elements that resist compression

(A) A high magnification view with sample **compression and tension elements** labeled to visualize the tensegrity force balance based on local compression and continuous tension.

(B) A schematic diagram of the complementary force balance between **tensed microfilaments (MFs)**, **intermediate filaments (IFs)**, **compressed microtubules (MTs)** and the ECM in a region of a cellular tensegrity array. **Compressive forces borne by microtubules** (top) are transferred to ECM adhesions when microtubules are disrupted (bottom), thereby increasing substrate traction.

Cellular Tensegrity Model



Microtubules, microfilaments and intermediate filaments within the cytoskeleton of endothelial cells visualized with GFP tubulin, rhodaminated phalloidin and antibodies to vimentin, respectively.

(A) Microtubules (**green**) span large regions of the cytoplasm and often appear curved in form.

(B) Microfilaments (**green-yellow**) appear linear in form within long stress fibers and triangulated actin 'geodomines'; blue staining indicates nuclei.

(C) Intermediate filaments (**red**) appear within a spread cell as a reticulated network that extends from the nucleus to the cell periphery.

Cytoskeleton: The skeleton of the cell

<https://youtu.be/j7jASIS0298?si=3Own1muCqwk-YDgb>

Bibliography

Biochemistry: Voet, Voet, Pratt

Biology- A global Approach: Campbell